

Real Numbers and Roots

34 points

Name _____

Date _____

Period _____

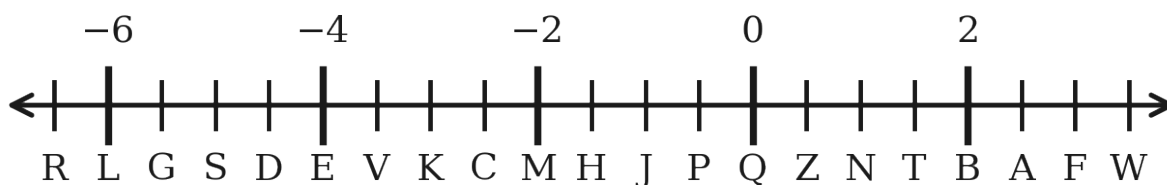
Score _____ / 34

*Show your work. Circle your final answer.***1. Locating Real Numbers on a Number Line**

7 points

1. A number line is shown. Give the letter of the point that is closest to each value.

3 points



a. $2 - \pi$

b. $\sqrt{6}$

c. $\sqrt[3]{9} - 8$

2. For each value, write the two consecutive integers it lies between, the integer it is closer to, and the comparison you used.

4 points

Value	Lies between	Closer to	How you know
a. $\sqrt{54}$			
b. $\sqrt[3]{100}$			
c. $-\sqrt{29}$			
d. 2π			

2. Roots and the Equations They Solve

11 points

3. Write every value that makes each statement true.

4 points

a. $\sqrt{121} = \underline{\hspace{2cm}}$

b. If $x^2 = 121$, then $x = \underline{\hspace{2cm}}$

c. $\sqrt[3]{-125} = \underline{\hspace{2cm}}$

d. If $x^3 = -125$, then $x = \underline{\hspace{2cm}}$

4. For each equation, select all of the solutions.

4 points

	$x = -4$	$x = 4$	$x = -8$	$x = 8$	no real solutions
a. $x^2 = 64$	☐	☐	☐	☐	☐
b. $x^3 = 64$	☐	☐	☐	☐	☐
c. $x^3 = -64$	☐	☐	☐	☐	☐
d. $x^2 = -64$	☐	☐	☐	☐	☐

5. Complete each statement.

3 points

a. When x is any real number other than 0 and it is multiplied by itself, the result will

☐ always ☐ sometimes ☐ never

be a

☐ positive ☐ negative

value.

b. Therefore, equations in the form $x^2 = p$, where p is a whole number **greater than 0**, will have

☐ only one solution. ☐ two solutions. ☐ three solutions.

c. How many solutions does $x^2 = 0$ have?

3. Ordering and Comparing Real Numbers

10 points

6. List the numbers below from least to greatest.

1 point

$$\frac{27}{4}, -3.2, -\sqrt[3]{60}, \sqrt{48}, 7.05$$

--	--	--	--	--

7. Compare each pair. Select <, > or =.

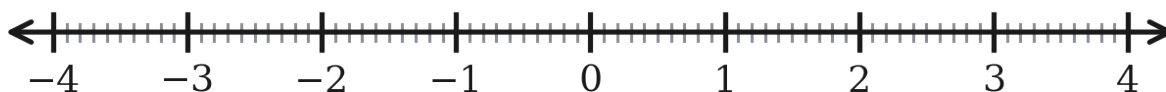
5 points

a.	$8 - \pi$	☐ < ☐ > ☐ =	$2(2.6)$
b.	$\sqrt{50}$	☐ < ☐ > ☐ =	7
c.	$-\sqrt{10}$	☐ < ☐ > ☐ =	-3.5
d.	$\sqrt[3]{-8} \cdot (-6)$	☐ < ☐ > ☐ =	$\sqrt{144}$
e.	$\sqrt{78}$	☐ < ☐ > ☐ =	3π

8. Plot and label each value on the number line.

4 points

$$-\sqrt[3]{50}, -\frac{11}{5}, \sqrt{3}, \pi$$



4. Rational or Irrational

6 points

9. Write each number in the correct column of the table.

6 points

$$\frac{\sqrt{81}}{3}, -0.1\sqrt{144}, \frac{57}{10}, -\sqrt[3]{9}, \sqrt{43}, -0.23\pi$$

Rational Number	Irrational Number
-----------------	-------------------

--	--